

MOCPSO: A multi-objective cooperative particle swarm optimization algorithm with dual search strategies[☆]

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ABSTRACT

Particle swarm optimization (PSO) is a widely embraced meta-heuristic approach to tackling the complexities of multi-objective optimization problems (MOPs), renowned for its simplicity and swift convergence. However, when faced with large-scale multi-objective optimization problems (LSMOPs), most PSOs suffer from limited local search capabilities and insufficient randomness. This can result in suboptimal results, particularly in high-dimensional spaces. To address these issues, this paper introduces MOCPSO, a Multi-Objective Cooperative Particle Swarm Optimization Algorithm with Dual Search Strategies. MOCPSO incorporates a diversity search strategy (DSS) to augment perturbation and enhance the local search scope of particles, alongside a more convergent search strategy (CSS) to expedite particle convergence. Moreover, MOCPSO utilizes a three-category framework to effectively leverage the benefits of both DSS and CSS. Experimental results on benchmark LSMOPs with 500, 1000, and 2000 decision variables demonstrate that MOCPSO outperforms existing state-of-the-art large-scale multi-objective evolutionary algorithms on most test instances.

1. Introduction

Multi-objective optimization problems (MOPs), i.e. optimization problems with multiple (conflicting) objectives, are very common in real-world applications, such as software engineering, machine learning, recommendation systems, and Financial Cloud Computing etc [1–6]. Different from one global optimal solution in single-objective optimization problem, due to the confliction of different objectives, each MOPs has a set of trade-off solutions [7]. Without loss of generality, a minimization MOP can be defined as:

$$\min \mathbf{F}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x}))^T \quad (1)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_D)$ is the decision vector with D decision variables, M is the number of functions, and $f_i(\mathbf{x})$ is the i th objective function. Roughly speaking, MOPs with more than 100 decision variables can be seen as large-scale multi-objective problems (LSMOPs) [8,9].

Solution $\mathbf{x}$ is said to dominate solution $\mathbf{y}$, denoted by $\mathbf{x} < \mathbf{y}$, if the following conditions hold:

$$\forall i : (f_i(\mathbf{x}) \leq f_i(\mathbf{y})), \exists j : (f_j(\mathbf{x}) < f_j(\mathbf{y})) \quad (2)$$

where i and $j = 1, 2, \dots, M$. If no other solution dominates $\mathbf{x}$, it is the Pareto optimal solution. The set of all these solutions is called the Pareto Set (PS) and the corresponding mapping of PS in objective space is termed the Pareto front (PF). The purpose of optimizing a MOP is to find an approximation set of PS with good convergence and diversity.

For the past few years, although multi-objective evolutionary algorithms (MOEAs) have proven to be effective in solving multi-objective optimization problems (MOPs) with a moderate number of decision variables (typically less than 100), their performance tends to deteriorate when dealing with large-scale multi-objective optimization problems (LSMOPs). In the case of LSMOPs, existing algorithms frequently struggle to efficiently explore the expansive solution space, making them prone to becoming trapped in personal optima [10, 11]. In recent years, a series of large-scale multi-objective optimization evolutionary algorithms (LSMOEAs) have been proposed to address LSMOPs [12]. According to how the decision variables are handled, LSMOEAs can be categorized into three categories:

- (1) **Decision space reduction.** These algorithms transform high dimensional decision space into low dimensional decision space

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- [13]. PCA-MOEA [14] used principal component analysis to reduce the dimension of decision variables. In order to reduce dimensionality, MOEA/PSL [15] used a restricted Boltzmann machine and a denoising autoencoder. In [16], a weighted optimization framework (WOF) was proposed to deal with LSMOPs by problem transformation for each solution. Qin et al. [17] proposed LMOEA-DS that specifies two weight variables to move a solution along two search directions for finding better solutions.
- (2) **Decision variable cluster.** The divide-and-conquer approach is commonly employed in algorithms designed to handle LSMOPs, often involving the grouping of decision variables [18]. CCGDE3 [19] was the first algorithm to propose a random grouping technique and a co-evolution framework to handle LSMOPs. In [20], MOEA/D was based on decision variable decomposition, and [21] achieved good results in experiments with 1200 decision variables. Ma et al. [9] proposed MOEA/DVA which divides decision variables by analyzing the relationship between their control properties and objective functions. MOEA/DVA divided the decision variables into position variables, distance variables, and mixed variables and optimizes them alternately. In [8], according to the angle between the fitted line and the convergence direction, LMEA [8] divides the decision variables into convergence correlation and diversity correlation. LMEA optimizes two types of variables alternately by different strategies and performs better than the MOEA/DVA. mogDG-shift [22] grouped variables by analyzing their attribute interactions and achieved good performance.
- (3) **Novel strategies.** Different effective strategies are used by these MOEAs to improve algorithm performance [23,24]. In [25], MCSO divided the population into three sub-populations and updated two losers by the same search strategy. Aiming to generate adaptive solutions, DGEA [26] proposed by He et al. generated solutions along the direction between non-dominant and dominant solutions. S3-CMA-ES [27] used a covariance matrix to evolve adaptively and generated offspring using a probabilistic model. MOEA-CSOD [28] proposed by Liang et al. trained the solution of the current population through distributional adversarial networks, and generated offspring.

The majority of variant algorithms of PSO that do not necessitate a reduction in the decision space or clustering of decision variables fall under the category of novel strategies. In [29], MMOPSO proposed two search strategies to update the velocity of each particle, which is respectively beneficial for the acceleration of convergence speed and the keeping of population diversity. However, these strategies solely focus on the velocity component and may exhibit limited effectiveness in high-dimensional spaces. Tian et al. [30] proposed LMOCSO based on the competitive swarm optimizer, which improved the search efficiency of solving LSMOPs by adding an acceleration term to the position updating strategy. The high dimensionality of LSMOPs poses a challenge for traditional PSO algorithms, as they struggle to efficiently explore the solution space. To address this issue, this paper proposes a novel approach called multi-objective cooperative particle swarm optimization (MOCPSO). Unlike MMOPSO and LMOCSO, MOCPSO proposed two new search strategies which are more beneficial to the convergence and diversity of the population, and a three-competition framework to distribute the search strategy reasonably. MOCPSO is specifically designed to overcome the limitations of PSO in high-dimensional LSMOPs, enabling more efficient search and optimization. The main contributions of this paper are summarized as follows:

- (1) In this paper, a Diversity Search Strategy (DSS) is introduced to enhance the diversity of particle swarm search. The DSS involves increasing the velocity of particles away from adjacent non-dominated particles and random particles, thereby promoting a more extensive search of the solution space. This strategy aims

to improve the overall search capability of the algorithm and increase the chances of discovering diverse and non-dominated solutions.

- (2) To enhance the search efficiency, this paper introduces a Convergence Search Strategy (CSS). The CSS incorporates a segment of displacement into the position updating strategy. By integrating this strategy, the algorithm is able to guide particles towards promising regions in the search space, facilitating convergence towards optimal solutions.
- (3) To verify the effectiveness of MOCPSO, this paper compares the performance of seven state-of-the-art MOEAs on benchmark problems. The compared algorithms include three popular PSO algorithms (MMOPSO [29], WOF-SMPPO [16], and LMOCSO [30]) and four well-known large-scale MOEAs (LMEA [8], DGEA [26], LMOEA-DS [17], and IM-MOEA-D [31]). The experimental results show the outstanding performance of the proposed MOCPSO.

The rest of the paper is organized as follows. Section 2 introduces the background knowledge and motivation of MOCPSO. In Section 3, the details of MOCPSO are demonstrated. Section 4 presents the experimental results on benchmark problems. Finally, the conclusions are drawn in Section 5.

2. Background and motivation

2.1. Definition of PSO

Particle swarm optimization is a well-known swarm intelligence paradigm originally inspired by social behavior (e.g., fish schooling and bird flocking [32]) [33]. In the traditional PSO framework, each solution is represented as a particle, which comprises a velocity vector and a position vector. These particles are evaluated using an evaluation function that determines their performance. Winners and losers are distinguished based on their evaluation results. The updating process involves a competitive strategy, where losers learn from winners. Specifically, the velocity vector of the loser is updated by considering its current velocity, as well as the velocities of personal winner particles and global winner particles. The position vector of the loser is then updated based on its current position and the updated velocity vector. This iterative process allows the particles to adapt and improve their positions in the search space over time. The updating process for the velocity and position in traditional PSO is as follows:

$$\begin{aligned} \mathbf{v}_l(t+1) &= c_1 \mathbf{v}_l(t) + c_2 r_1 (\mathbf{x}_{pw}(t) - \mathbf{x}_l(t)) + \\ &\quad c_3 r_2 (\mathbf{x}_{gw}(t) - \mathbf{x}_l(t)) \\ \mathbf{x}_l(t+1) &= \mathbf{x}_l(t) + \mathbf{v}_l(t+1) \end{aligned} \quad (3)$$

$$(4)$$

where $\mathbf{v}_l(t)$ represents the velocity of the loser and $\mathbf{x}_l(t)$ represents the position of the loser in the t -th iteration. $\mathbf{x}_{gw}(t)$ and $\mathbf{x}_{pw}(t)$ are the positions of the global optimal particle and the personal optimal particle, respectively. r_1 and r_2 are two independent random variables in the range $[0,1]$. c_1 , c_2 and c_3 are the constant that control the loser velocity, the personal and the global optimum particle influence, respectively.

2.2. Multi-objective PSO algorithms

Many excellent PSO algorithms have been proposed to cope with MOPs [34,35]. According to the view of handling the problems, this paper divides PSO algorithms into three categories and introduces them in the next.

The first category of PSO algorithms focuses on finding the global optimal solution and involves the application of velocity limits after updating the particles [36]. For example, in MO_Ring_PSO_SCD [37], the best global position is the position of the best particle within the particle's neighborhood determined by a ring topology. MOPSO [38]

and NMPSO [39], randomly pick the global optimal solution from the archive. In SMPSO [40], each velocity is multiplied by a contraction factor after the update, and constrained based on the boundary values of the decision variables.

The second category introduces a novel framework and search strategy aimed at enhancing the efficiency of PSO algorithms [41,42]. CCSMPSO [43] proposed by Atashpendar et al. enhances algorithm performance and parallelization through cooperative co-evolution. CMPSO [44], an effective method that integrates the principles of PSO with the Multi-Objective Particle Swarm Optimization (MPSO) technique, has been developed. It introduces a co-evolving multi-swarm framework and utilizes external shared profiles to facilitate the exchange of search information among different populations. LSTPA [45] divides solutions into convergence and diversity subpopulations, aiming for both convergence and diversity. To improve convergence speed, a fitness-aware variation operator (FAVO) is used to guide Diversity solutions towards Convergence. Additionally, an adaptive penalty based boundary intersection (APBI) strategy is employed for environmental selection, balancing convergence and diversity during different stages of the evolution process.

The third class of PSO algorithms adopts a novel approach to tackle challenging optimization problems [46]. RSPSO [47] is a novel algorithm proposed in this paper for optimization problems. The swarm is divided into elite and normal groups based on a ranking system. The search process is divided into four stages using a switching framework, with adaptive changes in learning strategies and parameter settings. In CPSO [48], it addresses the challenges of diversity and convergence by employing multiple swarms that coevolve in a distributed manner to cover different parts of the Pareto front. MPCoPSO [49] addresses the challenges of selecting stocks with limited cardinality while balancing returns and risks. MPCoPSO utilizes a hybrid encoding strategy, a return risk ratio heuristic, a bi-directional local search, and a hybrid elite competition strategy to improve solution accuracy and convergence speed. IM-MOEA-D [31] employs a novel approach to partition the objective space by utilizing k-means clustering. This grouping scheme helps in controlling the diversity of solutions generated. Additionally, the algorithm introduces a selection criterion called global replacement, which selects the most suitable reference vector from the entire population based on decomposition. In SDLPSO [50], it aims to balance swarm diversity and convergence speed while minimizing computational resources. The algorithm updates particles based on dominance relationships, where a particle is updated only when its two randomly selected exemplars from the swarm dominate it. ALMOEA [51] uses a trained neural network to learn a gradient-descent-like direction vector (GDV) for each solution. This GDV guides the generation of new solutions efficiently. The algorithm incorporates an MLP with one hidden layer and utilizes backpropagation for training.

2.3. Motivation

Traditional particle swarm optimization algorithms exhibit certain limitations and flaws. By extending formulas (3) and (4) to time $t=0$, the following formula is obtained:

$$\mathbf{v}_i(t+1) = c_1^t \mathbf{v}_i(1) + c_2 r_1 (\phi_{pw}) + c_3 r_2 (\psi_{gw}) \quad (5)$$

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(0) + \mathbf{v}_i(1) + \dots + \mathbf{v}_i(t+1) \quad (6)$$

where ϕ_{pw} and ψ_{gw} represent the accumulations of historical personal and global most guided vectors, respectively. In general, the initial velocity is zero and the initial position is random. So the velocity at $t=1$ can be thought of as random. According to formula (5), as the velocity of PSO increases with t , causing the specific gravity of $\mathbf{v}_i(1)$ to gradually decrease, while the specific gravity of ϕ_{pw} and ψ_{gw} gradually increases. Similarly, the position at $t+1$ is primarily determined by ϕ_{pw} and ψ_{gw} , with minimal influence from the initial position $\mathbf{x}_i(0)$. When confronted with high-dimensional LSMOPs, the reliance of PSO solely

on the personal best (pw) and global best (gw) positions can result in a limited exploration of the local search space.

Through the above analysis, the traditional PSO algorithm still faces the following problems when solving LSMOPs:

1. The traditional PSO algorithm lacks sufficient random exploration, making it prone to getting trapped in personal optimal solutions.
2. In high dimensional space, the convergence of PSO still has room for improvement.
3. In general, the PSO algorithm tends to prioritize the update of loser particles, which can result in a lack of active search for the winner particles. This imbalance in search can ultimately lead to reduced efficiency and suboptimal performance of the algorithm.

To address the aforementioned issues, this paper presents a more effective PSO algorithm called MOCPSO with Dual Search Strategies. Firstly, the algorithm incorporates a Diversity Search Strategy (DSS) that guides particles to approach nearby random particles while moving away from adjacent non-dominated particles. This strategy promotes exploration and diversification in the search space. Secondly, MOCPSO introduces a novel Convergence Search Strategy (CSS) that aims to facilitate the convergence towards optimal solutions. Furthermore, both DSS and CSS are designed to be independent of global and specific optimal particles. This approach ensures that the algorithm is not overly influenced by a small subset of particles and can explore the search space more comprehensively. Finally, the three-competition framework ensures that all particles are updated and effectively allocates particles using both DSS and CSS. This framework facilitates a balanced convergence and diversity trade-off, leading to improved algorithm performance.

3. The proposed approach

In this section, a comprehensive overview of the algorithm process employed by MOCPSO will be provided. MOCPSO primarily consists of two key components: DSS and CSS. To illustrate the particle behavior under different approaches, Fig. 1 presents a schematic diagram showcasing the displacement of particles when utilizing traditional PSO, DSS, and CSS, respectively. This diagram offers a visual representation of the distinct effects produced by each approach. Fig. 1(a) represents the initial state of traditional PSO, and the various types of arrows indicate different velocity directions for the particles. Fig. 1(d) demonstrates the search process and results obtained by traditional PSO. The red arrow represents Eq. (3) in traditional PSO, which guides the initial particle to move towards the final position after one update. This movement process corresponds to Eq. (4). The initial state of DSS is depicted in Fig. 1(b). In comparison to Fig. 1(a), DSS introduces two additional velocity components: the offset direction and the random direction. This allows the particles to explore a wider area during the optimization process. Fig. 1(e) represents the search process of the DSS. The red arrow indicates the actual displacement, which corresponds to Eq. (7). The inclusion of two additional directions enhances the search efficiency within the local space and promotes greater diversity among the particles, leading to improved overall performance. Fig. 1(c) depicts the initial configuration of CSS and Fig. 1(f) illustrates the process of CSS. Similar to traditional PSO, CSS employs the same movement strategy represented. However, CSS adds another displacement (blue arrow corresponding to Eq. (11)) based on this movement strategy. This displacement aids in bringing the particle closer to the guide particle, thereby enhancing the convergence of the algorithm when dealing with LSMOPs.

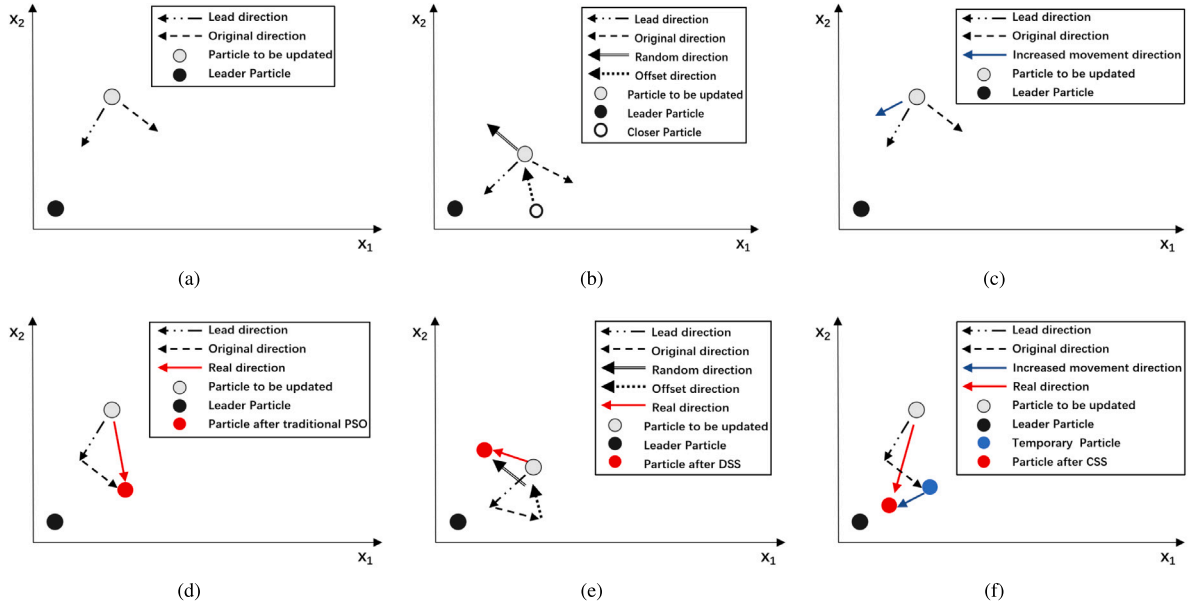


Fig. 1. The figure provides a schematic diagram of Eq. (3)–Eq. (11) to illustrate the optimization process in the t -th iteration. The diagrams denoted as 1(a), 1(b), and 1(c) illustrate the initial positions of the traditional Particle Swarm Optimization (PSO), Diversity Search Strategy (DSS), and Convergence Search Strategy (CSS) respectively. 1(d), 1(e) and 1(f) indicate the search process and result by traditional PSO, DSS and CSS respectively.

3.1. Diversity search strategy(DSS)

The diversity search strategy(DSS) can increase the exploration area of particles. It includes velocity search strategy Eq. (7), and the position search strategies Eq. (8). In DSS, particles converge not only to winner but also to other particles. In Eq. (7), two modifications are introduced to enhance exploration. Firstly, the algorithm selects the non-dominated solution that is closest to the loser particle as the reference point. A vector pointing away from this reference point is then added to the velocity rule, promoting exploration towards unexplored regions. Secondly, DSS incorporates a random position component, increasing uncertainty and enabling particles to explore a wider range of areas in the search space. These modifications aim to improve the algorithm's exploration capabilities and enhance its ability to discover diverse and potentially optimal solutions.

$$\mathbf{v}_i(t+1) = c_1 r_1 \mathbf{v}_i(t) + c_2 r_2 (\mathbf{x}_w(t) - \mathbf{x}_i(t)) + c_3 (\mathbf{x}_r(t) - \mathbf{x}_i(t)) + c_4 r_3 (\mathbf{x}_i(t) - \mathbf{x}_c(t)) \quad (7)$$

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1) \quad (8)$$

where $\mathbf{x}_i(t)$ and $\mathbf{x}_w(t)$ represent loser and winner from the three-competition framework, respectively, $\mathbf{x}_c(t)$ is the closest non-dominated solution to $\mathbf{x}_i(t)$ in the same sub-population, and $\mathbf{x}_r(t)$ is a random position whose range is based on the value of $\mathbf{x}_i(t)$. c_1 , c_2 , c_3 and c_4 are the parameters which control the proportion of different parts respectively. r_1 , r_2 and r_3 are random variables between 0 and 1. Fig. 1(e) shows the update process of loser particle by Eq. (7)–(8).

The pseudocode of DSS is given in Algorithm 1. Firstly, DSS uses the non-dominated sorting algorithm [52] to sort the loser and then calculates the Euclidean distances of the solutions to each other which is in the same non-dominated order. According to Euclidean distances, the closest particle $\mathbf{x}_c$ can be found. $\mathbf{x}_r$ is a random particle with a random range from $\alpha * \min P_{li}$ to $\beta * \max P_{li}$ where $\min P_{li}$ is the minimum value of the current loser dimension variables and $\max P_{li}$ is the maximum value. α is a constant value from 0 to 1 and β should be larger than 1. In this way, a random particle is likely to appear around the loser and the range of random particles is present in Fig. 2. Finally, the loser particle is updated and produces offspring.

Algorithm 1: Diversity Search Strategy(DSS)

Input : P_l (Loser particle), P_w (Winner particle)

Output: $Offspring_{dss}$

```

1  $Offspring_{dss} \leftarrow \emptyset$ ;
2  $SortIndex \leftarrow NDSort(P_l)$ ; //non-dominated sort
3 for  $i \leftarrow 1$  to  $size(P_l)$  do
4    $D_i \leftarrow$  calculate distances in the same non-dominated order according to  $SortIndex$ ;
5    $\mathbf{x}_c \leftarrow$  based on  $D_i$ , find non-dominated solution closet  $P_{li}$ ;
6    $\mathbf{x}_r \leftarrow$  each dimension is a random value ranging from  $\alpha * \min P_{li}$  to  $\beta * \max P_{li}$ ;
7    $CurOffspring_v \leftarrow$  Using  $P_{li}$ ,  $P_{wi}$ ,  $\mathbf{x}_c$  and  $\mathbf{x}_r$  to calculate the velocity by Eq. (7);
8    $CurOffspring \leftarrow$  Using  $CurOffspring_v$  and  $P_{li}$  to calculate the position by Eq. (8);
9    $Offspring_{dss} \leftarrow Offspring_{dss} \cup CurOffspring$ ;
10 end
11 return  $Offspring_{dss}$ 

```

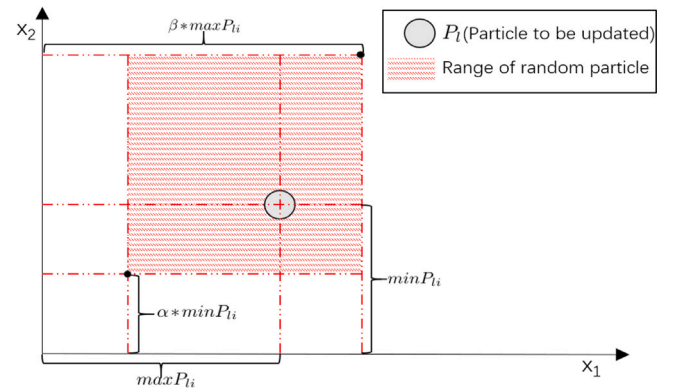


Fig. 2. A schematic diagram of the range of random particle. The red-shaded area is the range where random particles may appear. $\min P_{li}$ represents the minimum value of the decision variable of particle P_{li} and $\max P_{li}$ represents the maximum value.

3.2. Convergence search strategy(CSS)

In order to speed up particle convergence, this paper proposes a new search strategy which is called convergence search strategy(CSS) and showed in Algorithm 2. The velocity search strategy is Eq. (9), and the position search strategies are Eq. (10) and (11).

$$\mathbf{v}_l(t+1) = r_1 \mathbf{v}_l(t) + r_2(\mathbf{x}_w(t) - \mathbf{x}_l(t)) \quad (9)$$

$$\mathbf{x}_l(t) = \mathbf{x}_l(t) + \mathbf{v}_l(t+1) \quad (10)$$

$$\mathbf{x}_l(t+1) = \mathbf{x}_l(t) + c_1 r_3(\mathbf{x}_w(t) - \mathbf{x}_l(t)) \quad (11)$$

where $\mathbf{x}_l(t)$ and $\mathbf{x}_w(t)$ represent loser and winner in the three-competition framework, respectively, c_1 is the parameter and r_1 , r_2 and r_3 are three independent random variables between 0 and 1. Fig. 1(f) shows the update process of the loser particle by Eq. (9)–(11).

Algorithm 2: Convergence Search Strategy(CSS)

Input : P_l (Loser particle), P_w (Winner particle)
Output: $Offspring_{css}$

```

1  $Offspring_{css} \leftarrow \emptyset$ ;
2 for  $i \leftarrow 1$  to  $size(P_l)$  do
3    $CurOffspring_v =$  Using  $P_{li}$  and  $P_{wi}$  to calculate the velocity
   by Eq. (9);
4    $CurOffspring_p =$  Using  $P_{li}$  and  $CurOffspring_v$  to calculate
   the position by Eq. (10);
5    $CurOffspring =$  Using  $CurOffspring_p$  and  $P_{wi}$  to calculate
   the position by Eq. (11);
6    $Offspring_{css} \leftarrow Offspring_{css} \cup CurOffspring$ ;
7 end
8 return  $Offspring_{css}$ 
```

Compared to the traditional search strategy(Eq. (3) and (4)), the Convergence Search Strategy (CSS) in MOCPSO ensures that particles move closer to the winner particle, thereby avoiding getting trapped in personal optimization. Additionally, in Eq. (11), CSS further adjusts the position update by moving even closer to the winner particle. This adjustment is made to facilitate faster convergence towards optimal solutions. By incorporating these modifications, CSS enhances the algorithm's ability to balance convergence and diversity, leading to improved convergence speed and optimization performance.

3.3. The overall process of MOCPSO

As presented in Algorithm 3, MOCPSO has a very simple process, where the population is randomly initialized, and reference vectors are created. Then the main loop consists of three components: the three-competition framework which is presented in Algorithm 4, the polynomial mutation and the environmental selection strategy. In three-competition framework, each sub-population produces offspring by cooperating with others.

Algorithm 3: Procedure of MOCPSO

Input : population size N
Output: final population P

```

1  $P \leftarrow$  RandomInitPopulation( $N$ );
2  $W \leftarrow$  UniformReferenceVectors( $N$ );
3 while termination criterion not fulfilled do
4    $Offspring \leftarrow$  Three-competition Framework( $P$ ); // Alg4
5    $Offspring' \leftarrow$  PolynomialMutation( $Offspring$ );
6    $P \leftarrow$  EnvironmentalSelection( $Offspring' \cup P, W$ );
7 end
8 return  $P$ 
```

MOCPSO uses the same environment selection strategy as RVEA [53]. Given $Offspring' \cup P$ and reference vectors as inputs, the angle between particles and reference vectors is calculated and the particle which has the best angle-penalized distance (APD) [53] is selected as offspring.

A three-Competition Framework contributes to balancing convergence and diversity. The details in MOCPSO are presented in Algorithm 4.

$$I(\mathbf{x}) = \min_{\mathbf{y} \in P, \mathbf{y} \neq \mathbf{x}} \left\{ \sqrt{\sum_{1 \leq i \leq M} sd(f_i(\mathbf{x}), f_i(\mathbf{y}))^2} \right\} \quad (12)$$

$$sd(f_i(\mathbf{x}), f_i(\mathbf{y})) = \max\{0, f_i(\mathbf{y}) - f_i(\mathbf{x})\} \quad (13)$$

where M denotes the number of objectives and $f_i(\mathbf{x})$ denotes the i th objective value of $\mathbf{x}$.

Algorithm 4: Three-Competition Framework

Input : current population P
Output: $Offspring$

```

1  $P_{subw}, P_{subm}, P_{subl} \leftarrow \emptyset$ ;
2  $Score \leftarrow$  Calculate the fitness score in  $P$  by Eq. (12);
3 while  $|P| > 3$  do
4    $\{p, q, s\} \leftarrow$  Select three particles from  $P$  randomly;
   s.t.  $Score(s) \leq Score(q) \leq Score(p)$ 
5    $P \leftarrow P \setminus \{p, q, s\}$ ;
6    $P_{subw} \leftarrow P_{subw} \cup p$ ;  $P_{subm} \leftarrow P_{subm} \cup q$ ;  $P_{subl} \leftarrow P_{subl} \cup s$ ;
7 end
8  $Offspring_{css} \leftarrow CSS(P_{subl}, P_{subw})$ ; //Alg.2
9  $Offspring_{dss} \leftarrow DSS(P_{subw}, P_{subw})$ ; //Alg.1
10 if  $rand > P$  then
11    $Offspring_m \leftarrow CSS(P_{subm}, P_{subw})$ ; //Alg.2
12 else
13    $Offspring_m \leftarrow DSS(P_{subm}, P_{subw})$ ; //Alg.1
14 end
15  $Offspring \leftarrow Offspring_{css} \cup Offspring_{dss} \cup Offspring_m$ ;
16 return  $Offspring$ 
```

Firstly, the population is calculated fitness score by the shifted density estimation(SDE) [54] method. SDE is able to evaluate the convergence and diversity of solutions comprehensively and has been widely used in MOEAs [55] and it is defined as Eq. (12) and (13). Then according to SDE score, all particles are randomly divided into three sub-population: P_{subw} , P_{subm} , and P_{subl} . We consider P_{subl} which has the lowest SDE score needs to be converged faster, so taking P_{subw} as the leader of P_{subl} and updating P_{subl} by convergence search strategy(CSS). In order to enhance the active exploration of particles, the P_{subw} with the highest score is updated by diversity search strategy(DSS). Since P_{subw} is the best sub-population, it takes itself as the leader and only explores diversity without convergence guidance. For P_{subm} , it is difficult to determine its current state. Therefore, random values are adopted to control P_{subm} for the utilization of different strategies. Finally, the framework merges all offspring and returns.

4. Empirical studies

In this section, experimental studies are performed to examine the performance of MOCPSO.¹ Firstly, the overall settings of the experiment are introduced, including standard benchmark problems, performance metrics, and experiment parameter settings. Additionally, MOCPSO is compared with existing six state-of-the-art algorithms including three multi-objective PSO algorithms and three MOEAs. Finally, in order to verify the effectiveness of the two new strategies CSS and DSS, MOCPSO is further compared with the two variants.

¹ <https://github.com/ilog-ecnu/MOCPSO>

Table 1
Characteristics of the benchmark problems.

Problem	Shape	Separability
LSMOP1	Unimodal	Linear Fully Separable
LSMOP2	Mixed	Linear Partially Separable
LSMOP3	Multi-modal	Linear Mixed
LSMOP4	Mixed Linear	Mixed
LSMOP5	Unimodal	Convex Fully Separable
LSMOP6	Mixed	Convex Partially Separable
LSMOP7	Multi-modal	Convex Mixed
LSMOP8	Mixed	Convex Mixed
LSMOP9	Mixed	Disconnected Fully Separable

4.1. Experimental settings

In the experiment, the large-scale multi-objective test suite LSMOP [56] is used to evaluate the algorithms. LSMOP contains nine test problems: LSMOP1-9. The characteristics of the nine test problems are shown in Table 1. The number of objectives M is set to 2 and 3, the number of decision variables D is set to 500, 1000, and 2000 and the number of sub-components is set to 5.

The Inverted Generational Distance (IGD [57]), which is a widely used metric, is adopted to quantify the convergence and diversity of the approximate PFs obtained by MOCPSO and the peer algorithms. It is described as follows:

$$IGD(P, PF') = \frac{1}{|PF'|} \left(\sum_{i=1}^{|PF'|} distance(\mathbf{p}_i, P)^p \right)^{\frac{1}{p}} \quad (14)$$

where $P = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ represents the approximation point set in a true PF, $PF' = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_{|PF'|}\}$ is a subset of the true PF. The distance between a solution and a solution set is defined as the minimum distance between the solution and all solution set members in the objective space. Here p is set to 1. When computing the IGD value, the true PF has to be available in advance. Generally, the smaller the IGD value, the better the quality of the solution set obtained by an algorithm.

In order to validate the performance of MOCPSO, this paper employs seven algorithms for comparisons which can be divided two categories: multi-objective PSO algorithms, including MMOPSO [29], WOF-SMPSO [16] and LMOCSO [30], and MOEAs for LSMOPs, including LMEA [8], DGEA [26] LMOEA-DS [17] and IM-MOEA-D [31].

All the compared algorithms are implemented on PlatEMO [58]. The population size is 200 for 2-objective MOPs and 500 for 3-objective MOPs. The number of fitness evaluations for each run is set to $15000 \times D$ and each algorithm is run 20 times independently for each test instance. Other parameters are set based on the default values in PlatEMO.

For the proposed algorithm, MOCPSO, in the CSS, c_1 is set to be 0.125. In the DSS, c_1 , c_2 , c_3 and c_4 are 1.3, 1.2, 0.13, and 1.28, respectively. In order to expand the search range of random particles, the value of α is set to 0.1 and the value of β to 2. In addition to parameters in the rules, MOCPSO uses polynomial mutation [59] with the mutation probability of $1/D$.

Furthermore, the average performance score (APS) [60] is computed to reveal the overall running time of these algorithms. Given a certain test instance, the performance score of an algorithm is defined as the number of algorithms that significantly outperforms it. Herein, the averaged performance score across all problem instances is consolidated for each of these algorithms, considering problem instances with 500 and 1000 decision variables. A lower APS value signifies superior performance.

4.2. The results of comparative experiments

Table 2 lists the IGD values of six compared algorithms on 2-objective LSMOP1-9 with 500, 1000, and 2000 decision variables.

Moreover, the best result on each test instance is shown in a yellow background, and the mean and standard deviation (in brackets) of the IGD values are presented. The experimental results clearly demonstrate that MOCPSO outperforms the other MOEAs on the benchmark test problems, exhibiting better overall performance. Specifically, MOCPSO achieves better performance than the other algorithms on 17 out of 27 test instances. Table 2 further indicates that as the number of fitness evaluations increases, the performance of MOCPSO steadily improves, highlighting the effectiveness and efficiency of the proposed algorithm.

However, it is observed that MOCPSO does not perform well on the 2-objective LSMOP6 problem. This can be attributed to the complex nature of the Pareto front in LSMOP6, which presents a greater challenge compared to other test instances. MOCPSO may not have specific strategies tailored to effectively address complex and multi-modal problems. Fig. 3 illustrates the non-dominated solution sets associated with the median IGD values for the six compared MOEAs on 2-objective LSMOP2, LSMOP5, and LSMOP9, each with 1000 decision variables. It is evident that the proposed MOCPSO is capable of achieving a superior set of non-dominated solutions. Notably, on LSMOP9, which features a disconnected Pareto front, MOCPSO demonstrates the best results in both 2 and 3 objectives.

For 3-objective, Table 3 presents the mean and standard deviation of IGD values on LSMOP1-9, where the Wilcoxon ranks sum test [61] is adopted and the best mean for each test instance is highlighted with a yellow background. On all 27 test instances under consideration, the MOCPSO achieves statistically significantly better IGD values on 13 test instances, which is followed by MMOPSO gaining 2 best results, WOF-SMPSO gaining 3 best results, LMEA gaining 1 best result, DGEA gaining 2 best results, LMOEA-DS gaining 1 best result and IM-MOEA-D gaining 3 best result.

While MOCPSO generally demonstrates good performance across most test instances, it is evident that its performance significantly deteriorates on the 3-objective LSMOP3 and LSMOP7 problems. This can be attributed to the challenging nature of LSMOP3 and LSMOP7, which are large-scale MOPs with multi-modal landscapes and feature the most difficult separable and non-separable functions. Dealing with these complexities becomes more challenging for the 3-objective optimization scenario. Additionally, MOCPSO lacks a targeted approach specifically designed to handle multi-modal problems, further contributing to its difficulty in effectively addressing LSMOP3 and LSMOP7. Fig. 3 plots the non-dominated solutions with the median IGD values obtained by the six compared MOEAs on 3-objective LSMOP1, LSMOP8, and LSMOP9 with 1000 decision variables. As shown in Fig. 3, MOCPSO is capable of achieving a solution set with better convergence and diversity than the other six algorithms.

The proposed MOCPSO demonstrates superior performance in terms of running time compared to existing popular LSMOEAs on benchmark test problems. This can be observed in Fig. 4, where MOCPSO achieves better APS on both 2-objective and 3-objective problems.

4.3. Ablation study

In this section, the effectiveness of both the Convergence Search Strategy (CSS) and Diversity Search Strategy (DSS) on the performance of the proposed MOCPSO is analyzed and tested. To evaluate their impact, MOCPSO is compared with two variants: MOCPSO_A and MOCPSO_B. These variants incorporate different combinations of CSS and DSS, allowing us to assess the individual and combined effects of these strategies on the algorithm's performance. The definition of the aforementioned versions is presented in Table 4, wherein the traditional strategy is denoted by Eq. (3)–(4), CSS is represented by Eq. (9)–(11), and DSS is indicated by Eq. (7)–(8).

Fig. 5 plots the convergence profiles in terms of IGD values gained by MOCPSO and MOCPSO_A on 2-objective and 3-objective LSMOP1, LSMOP2 and LSMOP4 with 500 decision variables, averaged 20 independent runs. Fig. 6 plots the convergence profiles in terms of IGD values gained by MOCPSO and MOCPSO_B.

Table 2

IGD values of MMOPSO, WOF-SMPSO, LMOCSO, LMEA, DGEA, LMOEA-DS, IM-MOEA-D and MOCPSO on 2-objective LSMOP1-9, where the best result on each test instance is shown in yellow background. A smaller IGD value indicates better performance.

No.	Dec.	MMOPSO	WOF-SMPSO	LMOCSO	LMEA	DGEA	LMOEA-DS	IM-MOEA-D	MOCPSO
LSMOP1	500	1.3463e-2 (3.82e-3-)	8.2101e-2 (5.68e-2-)	7.1524e-3 (7.89e-4-)	4.6538e-2 (9.55e-2-)	6.8448e-3 (8.88e-4-)	1.2643e-1 (2.98e-3-)	1.9208e-2 (9.38e-3-)	3.2070e-3 (2.08e-4)
	1000	6.5527e-2 (2.98e-2-)	1.0757e-1 (5.79e-2-)	1.3316e-2 (1.44e-3-)	1.3644e-1 (2.74e-1-)	1.8536e-2 (2.67e-3-)	1.4707e-1 (1.64e-3-)	3.1436e-2 (1.92e-2-)	3.6767e-3 (2.89e-4)
	2000	1.1251e-1 (2.04e-3-)	9.9015e-2 (4.14e-2-)	2.5121e-2 (6.37e-3-)	9.1920e0 (4.24e+0-)	6.5130e-2 (7.44e-3-)	1.5888e-1 (9.60e-4-)	7.2156e-2 (1.99e-2-)	4.7885e-3 (5.30e-4)
LSMOP2	500	3.7923e-2 (7.32e-4-)	8.8927e-3 (1.96e-3-)	2.8371e-2 (1.77e-3-)	5.5420e-2 (9.19e-3-)	8.1499e-3 (2.71e-4-)	1.0073e-2 (2.61e-4-)	4.8688e-3 (2.54e-5+)	5.4218e-3 (6.34e-4)
	1000	2.3755e-2 (4.79e-4-)	5.6525e-3 (1.04e-3-)	1.6618e-2 (4.01e-4-)	3.2282e-2 (4.07e-3-)	5.3614e-3 (2.90e-4-)	6.4757e-3 (1.58e-4-)	3.4343e-3 (9.19e-5+)	3.6721e-3 (3.05e-4)
	2000	1.4332e-2 (2.02e-4-)	4.4775e-3 (6.90e-4-)	9.2264e-3 (1.53e-4-)	1.8317e-2 (1.86e-3-)	3.6187e-3 (1.09e-4-)	4.2036e-3 (8.73e-5-)	2.6785e-3 (6.27e-6-)	2.5433e-3 (9.78e-5)
LSMOP3	500	1.5139e+0 (4.01e-1-)	9.6229e-1 (3.04e-1-)	9.0448e-1 (3.33e-1-)	2.4538e+0 (4.15e+0-)	1.2296e+0 (2.32e-1-)	1.5115e+0 (2.29e-2-)	5.0064e-1 (4.46e-5+)	7.4715e-1 (7.63e-2)
	1000	1.6874e+0 (1.99e-1-)	1.2231e+0 (2.83e-1-)	1.6844e+0 (1.30e+0≈)	2.1543e+0 (2.00e+0-)	1.3460e+0 (3.29e-1-)	1.5479e+0 (4.68e-3-)	5.0030e-1 (2.33e-4+)	8.6026e-1 (1.52e-1)
	2000	1.5550e+0 (6.69e-2-)	1.3448e+0 (3.40e-1-)	1.9590e+0 (1.27e+0≈)	6.3275e+2 (4.72e+2-)	1.3911e+0 (2.52e-1-)	1.5634e+0 (2.32e-3-)	5.0003e-1 (1.66e-4+)	9.1477e-1 (9.09e-2)
LSMOP4	500	3.6114e-2 (6.92e-3-)	2.7060e-2 (3.22e-3-)	8.8678e-3 (4.89e-4-)	4.3876e-2 (1.76e-2-)	6.3190e-3 (3.77e-4-)	1.0623e-2 (7.88e-4-)	7.4831e-3 (2.15e-4-)	5.0880e-3 (6.84e-4)
	1000	4.1660e-2 (7.24e-3-)	1.7907e-2 (2.94e-3-)	8.2476e-3 (3.37e-4-)	2.4675e-2 (7.91e-3-)	5.1102e-3 (2.98e-4-)	1.4470e-2 (8.23e-4-)	5.5265e-3 (3.70e-5-)	3.2068e-3 (2.00e-4)
	2000	2.8855e-2 (6.31e-4-)	1.2726e-2 (2.80e-3-)	7.2191e-3 (2.29e-4-)	3.9360e-2 (8.92e-3-)	7.2712e-3 (1.13e-3-)	1.0310e-2 (2.85e-4-)	4.3030e-3 (2.61e-5-)	2.2987e-3 (1.12e-4)
LSMOP5	500	4.4538e-3 (9.31e-4-)	1.8285e-1 (1.12e-1-)	5.3596e-3 (6.96e-4-)	5.4962e-1 (1.70e-1-)	1.2099e-2 (1.35e-3-)	1.6804e-1 (4.41e-2-)	5.3259e-2 (1.74e-3-)	3.7711e-3 (3.45e-4)
	1000	7.3585e-3 (1.35e-3-)	1.6953e-1 (1.04e-1-)	2.8269e-2 (1.90e-2-)	4.0960e-1 (9.13e-2-)	1.8405e-2 (2.19e-3-)	1.8824e-1 (6.90e-4-)	6.0781e-2 (5.06e-3-)	4.0099e-3 (3.95e-4)
	2000	5.7345e-2 (1.45e-3-)	1.8561e-1 (9.16e-2-)	1.1327e-1 (1.51e-1-)	1.5964e+0 (5.25e+0-)	2.6113e-2 (1.46e-3-)	1.8383e-1 (3.45e-2-)	6.1378e-2 (1.59e-3-)	5.1731e-3 (4.97e-4)
LSMOP6	500	3.1913e-1 (5.13e-2+)	3.2453e-1 (5.26e-2+)	7.5909e-1 (3.47e-3+)	5.3416e-1 (2.13e-1≈)	1.7172e-1 (7.82e-2+)	1.8930e-1 (9.56e-3+)	6.1488e-1 (8.00e-2+)	7.8499e-1 (1.50e-1)
	1000	2.4043e-1 (5.28e-2+)	1.9651e-1 (5.05e-2+)	7.5522e-1 (1.34e-3-)	5.2462e-1 (2.58e-1+)	1.7339e-1 (1.85e-1+)	1.9336e-1 (1.06e-2+)	7.0836e-1 (3.45e-2+)	7.4438e-1 (5.91e-3)
	2000	2.1301e-1 (2.93e-2+)	1.7788e-1 (3.29e-3+)	7.5452e-1 (3.26e-3+)	5.5067e-1 (2.21e-1≈)	2.0331e-1 (2.37e-1+)	1.9136e-1 (1.69e-2+)	7.0856e-1 (8.15e-3+)	8.1991e-1 (2.30e-1)
LSMOP7	500	1.0256e+0 (3.53e-1-)	9.6288e-1 (9.70e-2-)	1.5562e+0 (4.21e-1-)	1.3567e+0 (2.56e-1-)	1.3111e+0 (2.24e-1-)	1.4894e+0 (8.08e-3-)	1.1036e+0 (3.75e-1≈)	7.6956e-1 (3.89e-2)
	1000	9.0740e-1 (2.90e-1≈)	1.0788e+0 (1.49e-1-)	1.3865e+0 (3.57e-1-)	1.2551e+0 (3.84e-1-)	1.3334e+0 (2.06e-1-)	1.5060e+0 (3.44e-3-)	1.1431e+0 (5.62e-1≈)	7.8954e-1 (1.07e-2)
	2000	9.0417e-1 (1.34e-1-)	9.0035e-1 (1.45e-1-)	9.3630e-1 (4.27e-1≈)	1.4032e+0 (2.55e-1-)	1.2977e+0 (1.82e-1-)	1.5127e+0 (7.73e-4-)	1.4301e+0 (1.13e-2-)	7.5869e-1 (1.09e-1)
LSMOP8	500	2.5858e-2 (2.04e-3-)	5.1325e-2 (2.44e-2-)	2.4948e-2 (1.34e-3-)	3.7243e-2 (1.08e-2-)	1.4774e-2 (3.53e-3-)	1.8659e-2 (8.91e-4-)	9.7200e-3 (7.43e-4≈)	9.5316e-3 (3.24e-3)
	1000	3.1197e-2 (1.46e-3-)	4.2431e-2 (1.63e-2-)	1.3303e-2 (7.68e-4-)	3.1208e-2 (1.63e-3-)	9.8511e-3 (8.09e-4-)	2.8284e-2 (2.56e-3-)	7.2325e-3 (3.68e-5≈)	7.8685e-3 (1.17e-3)
	2000	5.3511e-2 (1.68e-3-)	3.7448e-2 (9.86e-3-)	7.1333e-3 (3.20e-4-)	3.5168e-2 (2.77e-3-)	9.4934e-3 (3.85e-4-)	7.6841e-2 (7.93e-3-)	5.6234e-3 (4.32e-4-)	5.0484e-3 (4.47e-4)
LSMOP9	500	2.0289e-1 (5.32e-3-)	3.5940e-1 (1.93e-1-)	4.2655e-2 (1.51e-2-)	5.0847e-1 (2.06e-1-)	1.4198e-1 (9.97e-3-)	2.8394e-1 (1.21e-1-)	8.4365e-3 (4.32e-4+)	1.0585e-2 (1.59e-3)
	1000	9.4197e-2 (1.56e-3-)	3.5008e-1 (1.93e-1-)	2.6650e-2 (3.37e-3-)	4.3336e-1 (1.34e-1-)	1.0702e-1 (8.39e-2-)	1.9054e-1 (5.96e-2-)	8.9216e-3 (2.91e-4-)	6.0728e-3 (4.14e-4)
	2000	4.4614e-2 (5.84e-4-)	2.1269e-1 (2.22e-1-)	1.4510e-2 (1.70e-3-)	5.1092e+1 (2.17e+1-)	4.5175e-2 (1.75e-3-)	1.6681e-1 (4.54e-2-)	8.4142e-3 (4.60e-4-)	4.3446e-3 (1.13e-4)
+/-/≈		3/23/1	3/24/0	2/22/3	1/24/2	3/24/0	3/24/0	9/14/4	

'+', '-' and '≈' indicate that the result is significantly better, significantly worse and statistically similar to that of MOCPSO, respectively.

In Fig. 5, it is evident that MOCPSO converges faster and achieves better performance compared to MOCPSO_A. This observation suggests that the Convergence Search Strategy (CSS) plays a significant role in enhancing the performance of MOCPSO. CSS ensures that P_{subl} approaches P_{subw} as closely as possible, facilitating the convergence of the

population towards the approximated Pareto fronts. This closer alignment between P_{subl} and P_{subw} contributes to the improved convergence and overall performance of MOCPSO.

From Fig. 6, it is evident that the Diversity Search Strategy (DSS) plays a crucial role in enhancing the performance of MOCPSO, as

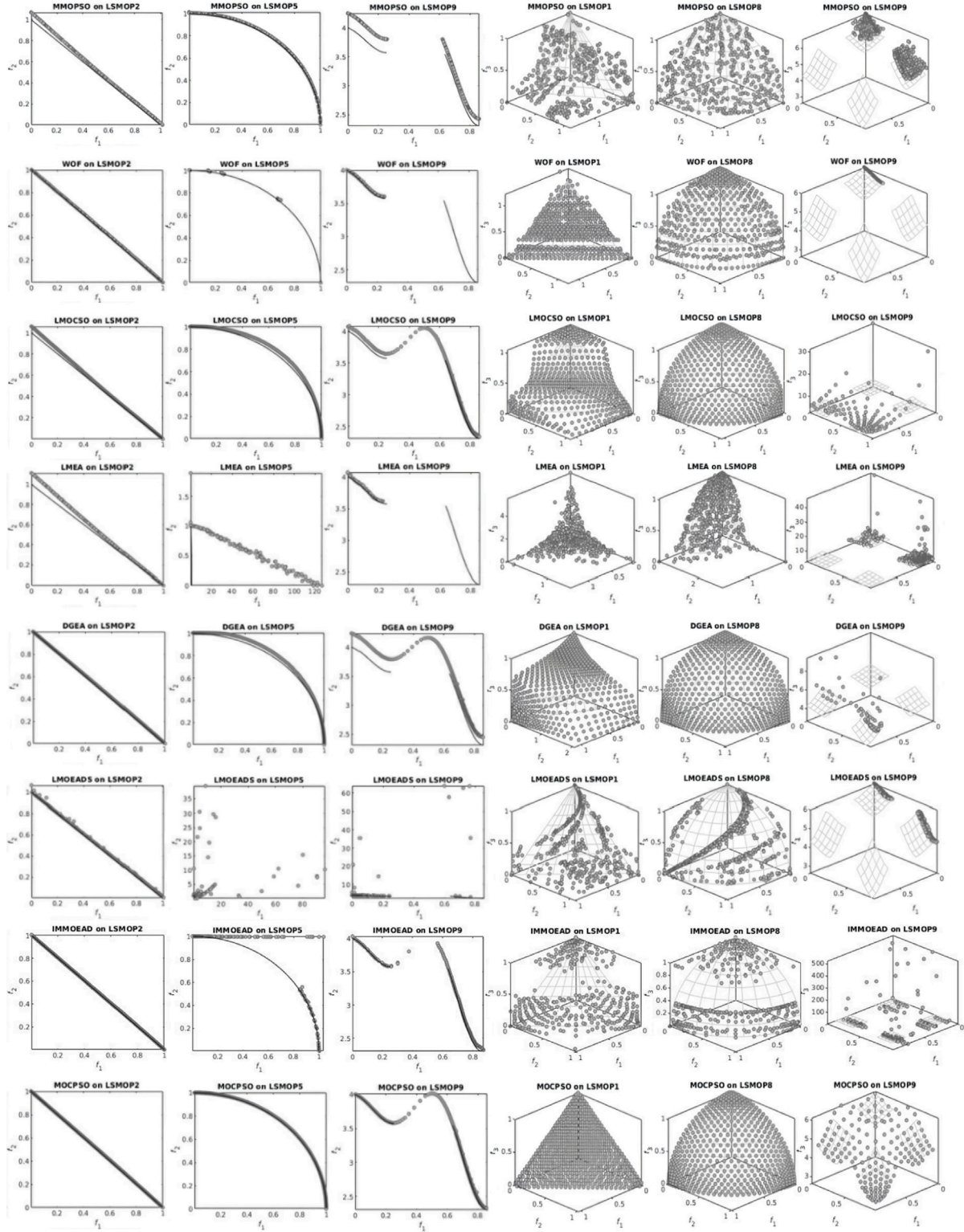


Fig. 3. Non-dominated solutions with the median IGD values obtained by MMOPSO, WOF, LMOCPSO, LMEA, DGEA, LMOEADS, IMMOEAD and MOCPSO with 1000 decision variables. The left three columns are 2-objective LSMOP2, LSMOP5, and LSMOP9 and the right three columns are 3-objective LSMOP1, LSMOP8, and LSMOP9.

observed when comparing MOCPSO with MOCPSO_B. DSS allows particles to explore a broader range of areas, leading to the generation of better solutions. This improved exploration capability significantly enhances the performance of the population in terms of convergence and diversity. The inclusion of DSS in MOCPSO contributes to its ability to achieve superior results compared to MOCPSO_B.

4.4. Parameter analysis

The parameter P plays a crucial role in the diversity search strategy, as different selections can influence the algorithm's performance. Table 5 presents the probability values (P) in various variants of MOCPSO. This paper conducts a comparative analysis of these variants on test sets LSMOP1, LSMOP2, and LSMOP8, considering different

Table 3

IGD values of MMOPSO, WOF-SMPSO, LMOCSO, LMEA, DGEA, LMOEA-DS, IM-MOEA-D and MOCPSO on 3-objective LSMOP1-9, where the best result on each test instance is shown in yellow background. A smaller IGD value indicates better performance.

No.	Dec.	MMOPSO	WOF-SMPSO	LMOCSO	LMEA	DGEA	LMOEA-DS	IM-MOEA-D	MOCPSO
LSMOP1	500	1.7183e-1 (1.04e-2-)	5.8727e-2 (9.46e-3-)	1.2716e-1 (6.88e-3-)	6.6695e-2 (5.32e-2-)	3.0542e-1 (7.89e-3-)	2.6967e-1 (3.50e-2-)	3.1156e-2 (8.09e-5-)	3.8740e-2 (8.65e-3)
	1000	2.4483e-1 (2.90e-2-)	6.0072e-2 (6.09e-3-)	1.3728e-1 (3.30e-3-)	9.7087e-2 (1.01e-1-)	3.2170e-1 (1.38e-2-)	2.7684e-1 (3.64e-2-)	2.8572e-1 (7.32e-3-)	4.4880e-2 (1.78e-2)
	2000	3.4344e-1 (1.40e-2-)	7.3672e-2 (1.28e-2-)	1.4196e-1 (1.77e-3-)	1.0118e+1 (3.28e+0-)	3.3582e-1 (1.70e-2-)	3.0710e-1 (1.70e-2-)	3.6349e-1 (1.06e-3-)	4.1857e-2 (1.16e-2)
LSMOP2	500	5.7016e-2 (1.27e-3-)	3.1412e-2 (1.21e-3-)	3.1681e-2 (2.63e-4-)	5.6832e-2 (6.06e-2-)	2.4945e-2 (7.72e-4-)	3.0269e-2 (5.98e-4-)	1.9840e-1 (3.29e-3-)	2.3779e-2 (8.21e-4)
	1000	3.8720e-2 (5.59e-4-)	2.2729e-2 (1.05e-3-)	2.2811e-2 (1.02e-4-)	4.7157e-2 (7.40e-2-)	1.9797e-2 (3.24e-4-)	2.4417e-2 (4.15e-4-)	2.6637e-2 (1.49e-5-)	1.9447e-2 (2.65e-4)
	2000	3.0017e-2 (5.07e-4-)	1.8570e-2 (4.90e-4-)	1.8892e-2 (4.54e-5-)	5.6675e-2 (1.02e-1-)	1.7697e-2 (1.27e-4≈)	2.1382e-2 (3.96e-4-)	2.4572e-2 (3.31e-5-)	1.7684e-2 (1.24e-4)
LSMOP3	500	5.2978e-1 (6.59e-2+)	6.0541e-1 (2.11e-1+)	1.8971e+0 (1.37e+0-)	8.9747e-1 (7.43e-2+)	5.5586e-1 (3.47e-2+)	5.1502e-1 (7.73e-2+)	5.0240e-1 (5.62e-3+)	1.0704e+0 (5.36e-1)
	1000	5.7427e-1 (9.15e-3+)	5.4851e-1 (2.09e-1+)	2.0317e+0 (1.26e+0-)	3.5267e+0 (2.86e+0-)	5.5570e-1 (2.16e-2+)	7.3997e-1 (1.06e-1+)	5.3061e-1 (3.05e-3+)	7.9929e-1 (4.91e-2)
	2000	5.8383e-1 (1.46e-2+)	5.0083e-1 (1.67e-1+)	2.9441e+0 (1.15e+0-)	7.5953e+1 (8.96e+1-)	6.2847e-1 (1.12e-2+)	7.7446e-1 (1.98e-2-)	5.0763e-1 (3.21e-3+)	7.3103e-1 (3.91e-2)
LSMOP4	500	1.6510e-1 (6.12e-3-)	7.0568e-2 (5.01e-3-)	8.4333e-2 (4.03e-3-)	6.4202e-2 (5.73e-2-)	7.8960e-2 (2.48e-3-)	8.0875e-2 (2.25e-3-)	4.4167e-2 (6.51e-4+)	4.9128e-2 (2.30e-3)
	1000	1.0391e-1 (3.04e-3-)	4.5556e-2 (3.74e-3-)	5.6033e-2 (1.53e-3-)	6.2879e-2 (7.41e-2-)	4.4749e-2 (1.10e-3-)	5.1770e-2 (7.17e-4-)	3.2422e-2 (5.69e-5+)	3.4634e-2 (1.35e-3)
	2000	6.1991e-2 (8.52e-4-)	2.9630e-2 (1.44e-3-)	3.5126e-2 (6.22e-4-)	6.2283e-2 (5.99e-3-)	2.7628e-2 (4.43e-4-)	3.4754e-2 (4.60e-4-)	2.6463e-2 (3.20e-5-)	2.4354e-2 (8.33e-4)
LSMOP5	500	2.2883e-1 (6.27e-3-)	2.6083e-1 (1.43e-1≈)	4.9081e-1 (4.16e-2-)	5.8568e+0 (4.18e+0-)	2.4846e-1 (7.19e-2-)	2.7361e-1 (1.16e-2-)	2.0356e-1 (9.05e-3-)	1.3692e-1 (1.59e-2)
	1000	2.3864e-1 (8.33e-3-)	1.5146e-1 (1.12e-1+)	5.2463e-1 (2.62e-2-)	6.9549e+0 (4.99e+0-)	2.3573e-1 (6.68e-3-)	2.7843e-1 (1.24e-2-)	2.1229e-1 (1.63e-3-)	1.6085e-1 (2.53e-2)
	2000	2.3900e-1 (6.11e-3-)	9.9850e-2 (6.59e-2+)	6.5148e-1 (1.75e-1-)	7.7385e+0 (5.47e+0-)	2.3565e-1 (5.76e-3-)	2.9101e-1 (8.31e-3-)	2.1700e-1 (2.90e-3-)	1.7041e-1 (2.39e-2)
LSMOP6	500	1.1078e+0 (2.10e-1≈)	1.1192e+0 (2.15e-1≈)	1.3397e+0 (2.92e-1-)	1.0584e+3 (2.72e+3-)	6.0976e-1 (1.51e-1+)	7.2821e-1 (1.26e-1+)	1.4515e+0 (1.07e-1-)	1.1731e+0 (2.30e-1)
	1000	7.9062e-1 (1.67e-1+)	1.2786e+0 (2.08e-3≈)	1.4074e+0 (3.31e-1≈)	3.4305e+3 (7.13e+3-)	5.7344e-1 (1.42e-2+)	7.0911e-1 (1.36e-1+)	1.1351e+0 (1.74e-1≈)	1.1744e+0 (1.59e-1)
	2000	6.5045e-1 (1.60e-1+)	1.2788e+0 (1.14e-1-)	1.3513e+0 (3.37e-1≈)	2.9017e+3 (7.37e+3-)	6.7219e-1 (3.03e-1+)	6.7571e-1 (2.59e-2+)	1.6461e+0 (3.84e-3-)	1.1707e+0 (1.48e-1)
LSMOP7	500	8.3965e-1 (1.60e-1≈)	8.1775e-1 (1.96e-3+)	9.4593e-1 (5.87e-6+)	9.4332e-1 (7.45e-2+)	7.9032e-1 (1.02e-1+)	8.3635e-1 (3.58e-2+)	8.9536e-1 (1.28e-2+)	9.4594e-1 (9.38e-6)
	1000	7.7490e-1 (1.68e-1+)	7.8295e-1 (1.21e-1+)	9.4569e-1 (9.51e-4+)	6.5823e-1 (1.12e-1+)	7.7159e-1 (6.76e-2+)	8.4059e-1 (1.64e-3+)	7.5963e-1 (2.11e-1+)	9.4599e-1 (4.77e-4)
	2000	4.8577e-1 (2.09e-1+)	5.9288e-1 (2.00e-1+)	9.4581e-1 (4.92e-4-)	1.4420e+1 (6.19e+1-)	7.2953e-1 (7.83e-2+)	8.2814e-1 (2.34e-2+)	9.1573e-1 (3.42e-2≈)	9.3537e-1 (1.08e-2)
LSMOP8	500	7.6618e-2 (2.36e-3-)	5.8315e-2 (2.76e-2-)	7.0645e-2 (1.94e-3-)	1.1072e-1 (1.04e-2-)	5.6965e-2 (4.70e-3-)	9.8003e-2 (5.14e-2-)	9.0805e-2 (2.03e-3-)	3.5082e-2 (3.44e-3)
	1000	6.5298e-2 (1.08e-3-)	5.6633e-2 (2.26e-2-)	5.5271e-2 (2.40e-3-)	7.3738e-2 (6.61e-3-)	5.4918e-2 (4.54e-3-)	8.0883e-2 (4.32e-3-)	1.0313e-1 (6.72e-3-)	2.9187e-2 (1.65e-3)
	2000	6.4073e-2 (1.61e-3-)	4.3181e-2 (7.07e-3-)	5.9081e-2 (7.25e-3-)	1.0148e-1 (1.15e-1-)	5.6931e-2 (4.14e-3-)	8.3888e-2 (3.55e-3-)	1.0632e-1 (7.31e-3-)	2.5720e-2 (5.26e-4)
LSMOP9	500	3.9763e-1 (1.70e-2-)	1.0341e+0 (2.28e-1-)	3.3409e-1 (4.36e-2-)	5.2557e-1 (2.99e-1-)	6.4410e-1 (1.06e-1-)	5.7089e-1 (1.83e-3-)	1.0034e-1 (6.63e-3≈)	1.5434e-1 (1.03e-1)
	1000	6.1587e-1 (7.85e-3-)	1.0501e+0 (2.05e-1-)	4.5686e-1 (5.99e-2-)	4.8724e-1 (2.72e-1-)	5.6745e-1 (1.22e-1-)	5.7029e-1 (1.13e-3-)	2.3022e-1 (1.41e-1-)	7.3388e-2 (1.16e-2)
	2000	3.0417e+0 (9.09e-1-)	9.1347e-1 (2.82e-1-)	7.5913e-1 (1.80e-1-)	1.1700e+2 (5.92e+1-)	5.7662e-1 (9.80e-2-)	5.6968e-1 (1.13e-3-)	9.2944e-2 (5.87e-3-)	6.2527e-2 (3.23e-3)
+/-/≈		7/18/2	8/16/3	2/23/2	3/24/0	9/17/1	8/19/0	7/17/3	

'+', '-' and '≈' indicate that the result is significantly better, significantly worse and statistically similar to that of MOCPSO, respectively.

dimensions (D) of 200 and 500, as well as varying objectives (M) of 2 and 3. The results, as shown in Table 6, indicate that MOCPSO_F performs notably better in terms of variants MOCPSO_C, MOCPSO_D, and MOCPSO_E. Overall, based on our analysis, we believe that P=0.35 is a suitable choice for the threshold parameter.

5. Conclusion and future work

In summary, this paper presents MOCPSO, a multi-objective cooperative particle swarm optimization algorithm that incorporates two

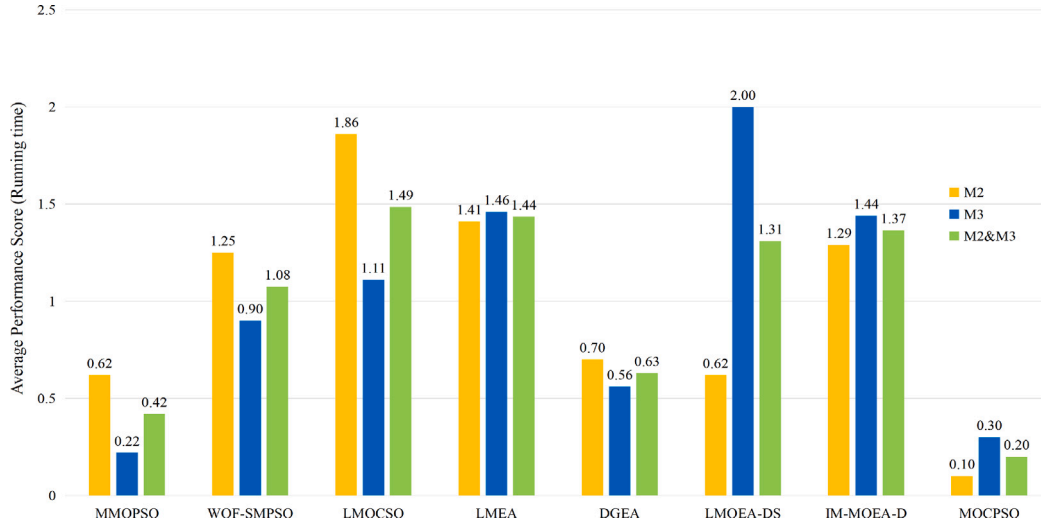


Fig. 4. The average performance score (APS) of Running time. The APS of an algorithm corresponds to the average number of algorithms that significantly outperforms it over all test instances. Thus a smaller APS value indicates better performance. As can be observed from the figure, the proposed MOCPSO achieves the better performance (with shortest bars) on problems with both 2 and 3 objectives.

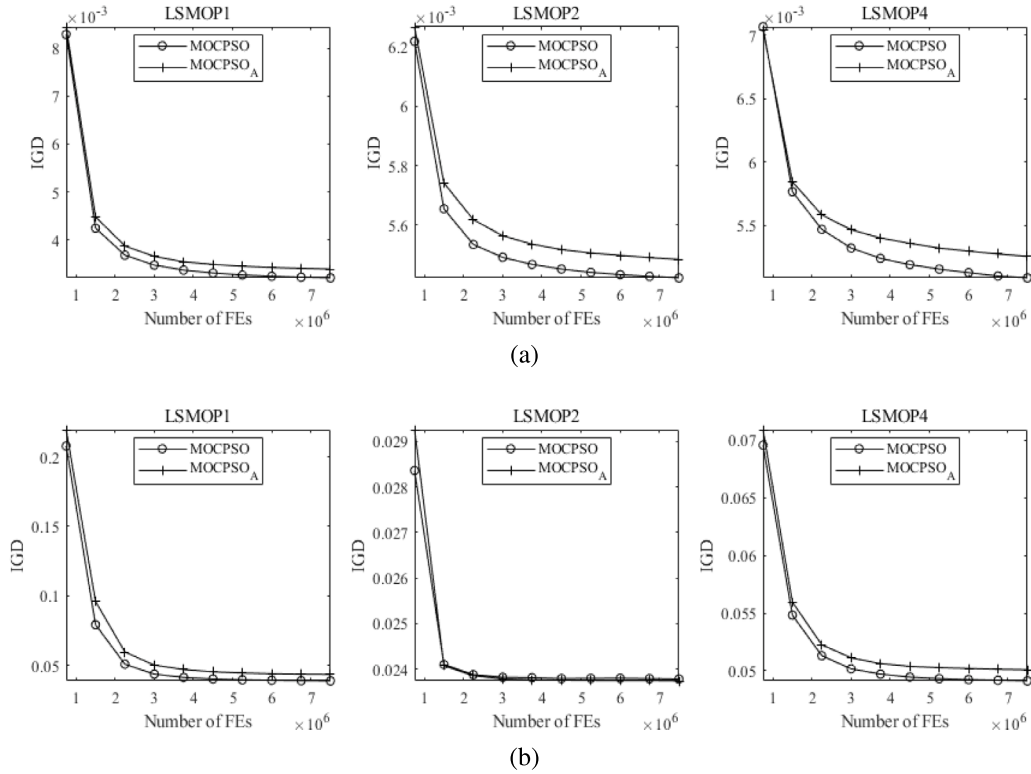


Fig. 5. Convergence profiles in terms of IGD values of MOCPSO and MOCPSO_A (without CSS) on LSMOP1, LSMOP2 and LSMOP4 with 500 decision variables, respectively. 5(a) is on 2-objective. 5(b) is on 3-objective.

Table 4

The definition of MOCPSO variants in ablation study.

Name	First search strategy	Second search strategy
MOCPSO	Diversity Search Strategy (Eq. (7)–(8), Fig. 1(e))	Convergence Search Strategy (Eq. (9)–(11), Fig. 1(f))
MOCPSO _A	Diversity Search Strategy (Eq. (7)–(8), Fig. 1(e))	Traditional PSO (Eq. (3)–(4), Fig. 1(d))
MOCPSO _B	Traditional PSO (Eq. (3)–(4), Fig. 1(d))	Convergence Search Strategy (Eq. (9)–(11), Fig. 1(f))

distinct search strategies: diversity search strategy (DSS) and convergence search strategy (CSS). DSS and CSS are characterized by their

unique characteristics, enabling MOCPSO to effective convergence and diversity In addition, MOCPSO employs a three-competition framework

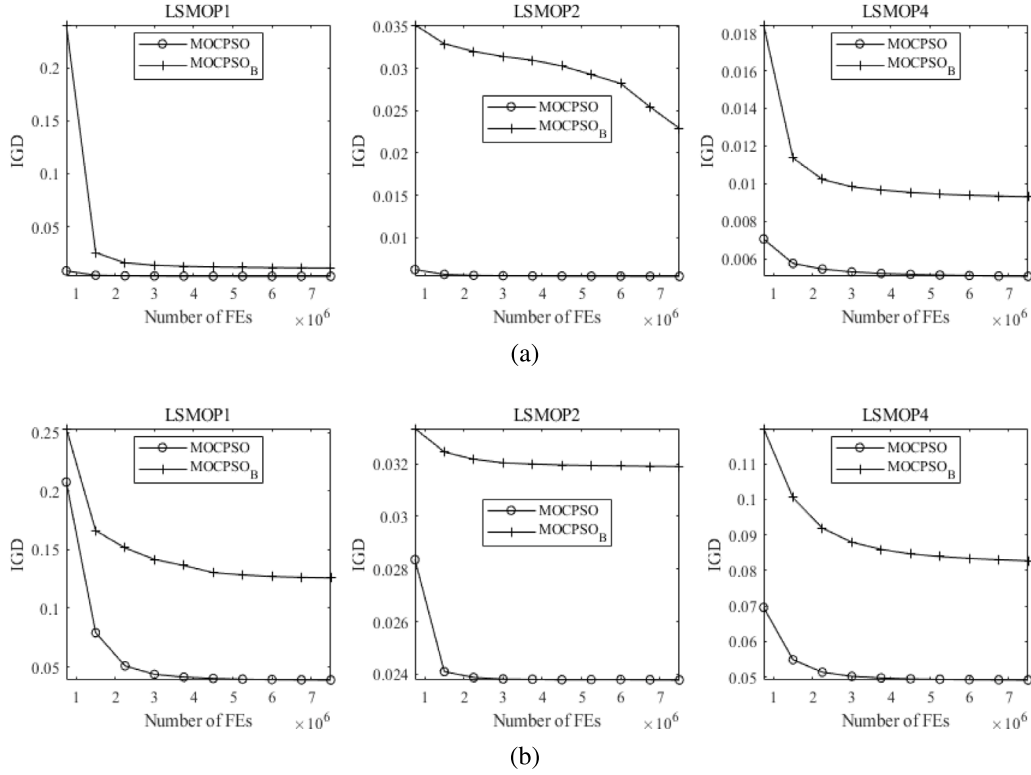


Fig. 6. Convergence profiles in terms of IGD values of MOCPSO and MOCPSO_B (without DSS) on LSMOP1, LSMOP2 and LSMOP4 with 500 decision variables, respectively. 6(a) is on 2-objective. 6(b) is on 3-objective.

Table 5
The probability P values in different variants of MOCPSO.

Variants	C	D	E	F
P	0.25	0.6	0.8	0.35

to partition the population into sub-populations, thereby enhancing its ability to maintain a balanced performance. Experimental results on benchmark test problems with up to 2000 decision variables involving seven large-scale multi-objective evolutionary algorithms demonstrate the competitive performance of the proposed MOCPSO.

Moving forward, future work will focus on enhancing the performance of MOCPSO, particularly in addressing multi-modal problems with complex landscapes like LSMOP3 and LSMOP7. We will develop strategies to effectively handle the challenges posed by these domains, improving the overall performance of MOCPSO. Additionally, we will strengthen our research on difficult-to-approximate (DtA) Pareto front boundaries and explore dynamic resource allocation to further enhance the method's performance [62]. Furthermore, we recognize the potential of applying MOCPSO to real-world problems such as community detection and neural network compression and will conduct research to adapt and optimize MOCPSO for these applications.

Table 6

IGD values across different variants of MOCPSO on 2/3 objective LSMOP1, LSMOP2 and LSMOP8, where the best result on each test instance is shown in yellow background. A smaller IGD value indicates better performance.

No.	M	Dec.	MOCPSO _C	MOCPSO _D	MOCPSO _E	MOCPSO _F
LSMOP1	2	200	2.8634e-3 (1.74e-4) ≈	3.3691e-3 (3.39e-4) –	3.5568e-3 (3.38e-4) –	2.8705e-3 (6.56e-5)
		500	3.2668e-3 (2.27e-4) ≈	3.8606e-3 (6.48e-4) –	4.4188e-3 (8.74e-4) –	3.2070e-3 (2.08e-4)
	3	200	4.0060e-2 (1.38e-2) ≈	3.8838e-2 (8.13e-3) ≈	5.5355e-2 (2.14e-2) –	4.1913e-2 (1.24e-2)
		500	4.5594e-2 (1.03e-2) –	5.7466e-2 (2.01e-2) –	6.0427e-2 (2.35e-2) –	3.8740e-2 (8.65e-3)
LSMOP2	2	200	8.7571e-3 (1.65e-3) ≈	9.0965e-3 (1.65e-3) ≈	9.5168e-3 (1.45e-3) –	8.5315e-3 (1.18e-3)
		500	5.3136e-3 (4.44e-4) ≈	5.5557e-3 (4.83e-4) ≈	5.6102e-3 (4.39e-4) –	5.4218e-3 (6.34e-4)
	3	200	3.9060e-2 (3.14e-3) ≈	4.0106e-2 (2.80e-3) ≈	3.9964e-2 (2.38e-3) ≈	3.8631e-2 (2.47e-3)
		500	2.4191e-2 (9.79e-4) ≈	2.4166e-2 (7.60e-4) ≈	2.4521e-2 (7.35e-4) –	2.3779e-2 (8.21e-4)
LSMOP8	2	200	1.3593e-2 (6.98e-3) ≈	1.4299e-2 (8.19e-3) ≈	1.7084e-2 (1.14e-2) ≈	1.3186e-2 (6.76e-3)
		500	1.0928e-2 (3.65e-3) ≈	9.7908e-3 (2.91e-3) ≈	9.8622e-3 (3.15e-3) ≈	9.5316e-3 (3.24e-3)
	3	200	5.0861e-2 (1.30e-2) ≈	4.3720e-2 (8.26e-3) ≈	5.3241e-2 (1.32e-2) ≈	4.8403e-2 (1.21e-2)
		500	3.5027e-2 (3.25e-3) ≈	3.5547e-2 (5.13e-3) ≈	3.7575e-2 (4.46e-3) ≈	3.5082e-2 (3.44e-3)
+/- / ≈			0/1/11	0/3/9	0/7/5	

'+', '-', and '≈' indicate that the result is significantly better, significantly worse, and statistically similar to that of MOCPSO, respectively.

CRediT authorship contribution statement

Yan Zhang: Methodology, Investigation, Formal Analysis, Writing – original draft. **Bingdong Li:** Methodology, Investigation, Formal Analysis, Writing. **Wenjing Hong:** Analysis, Writing – review & editing. **Aimin Zhou:** Analysis, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

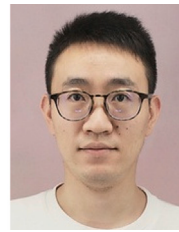
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